

Autoimmune Diseases

One by one: psoriasis, rheumatoid arthritis, lupus, Hashimoto thyroiditis, type 1 diabetes, celiac, multiple sclerosis, vitiligo and spondylitis (HLA-B27)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The seven most effective HLA markers are all negative: HLA-B*27, HLA-C*06:02 (2 markers), DRB1*15:01, DQ2.5/DR3, DQ8 and the shared epitope of DRB1*04. **PTPN22 R620W** is absent (G/G), and STAT4, IRF5, SH2B3 and ERAP1 do not have risk alleles. Therefore, diseases with strong HLA are low. Rheumatoid arthritis is at the 2.4 and 5.4 percentiles (seropositive, genomic). The psoriasis is at 6.3 and 8.2, with loci at 19.0. Celiac disease is at 3.9 and, without DQ2.5 or DQ8, it is practically excluded. The diabetes type 1 has the loci load at 8.5, and multiple sclerosis is at 23.9 and 25.1.

Unfavourable. Vitiligo is the highest signal: score at percentile 89.9, with **TYR R402 G/G** (2 copies of normal tyrosinase, the target of autoimmunity; it is the same allele that protects from melanoma). The vitiligo loci load is at 74.2. There are risk alleles in shared genes: **IL2RA T/T** and **IL7R C/C** (2 copies each; multiple sclerosis and diabetes type 1), **IFIH1 T/T**, and 1 copy in CTLA4 CT60, TNFAIP3, ITGAM R77H, INS and IL2/IL21. The lupus is in the 78.7 in the partial coverage score (37%), but in the 27.3 in the complete score and with loci in the 49.4. The ankylosing spondylitis is at 73.1, without B*27.

What does it mean? The overall autoimmune profile is low to typical, because HLA is missing, which is the major genetic factor of these diseases. The risk alleles that exist are in small effect genes (~1.1-1.3× each). The exception is vitiligo, with high polygenic risk. The values with the imputation replace those of §18 (which used kNN inference): psoriasis 8.7 → 6.3, celiac 12.5 → 3.9, lupus 76.5 → 78.7 and vitiligo 86.1 → 89.9. The autoimmune thyroid is detailed in §108 and sarcoidosis in §18.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality DR2 ≥ 0.8 . Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 23 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 22 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 2 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 18 PGS Catalog scores calculated locally with 2,590,552 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2547 loci, 2269 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk. The scores marked as already used in §18 were calculated with kNN inference; here they are recalculated with the imputation, and these values replace the previous ones.

Psoriasis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Psoriasis	PGS002293 (Shen 2022, 62 variants)	61 62 (weighted coverage 99%)	6.3	LESS LIKELY	favourable
Psoriasis	PGS005307 (Saklatvala 2025, 64 variants)	65 65 (weighted coverage 100%)	8.2	LESS LIKELY	favourable

- **Psoriasis:** In §18, 8.7 with kNN inference.

Rheumatoid arthritis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Rheumatoid arthritis	PGS004326 (Zhang 2023, 154 variants)	137 154 (weighted coverage 86%)	2.4	LESS LIKELY	favourable
Rheumatoid arthritis	PGS004118 (Monti 2024, 91 variants)	89 91 (weighted coverage 97%)	39.2	TYPICAL	typical
Seropositive rheumatoid arthritis (genomic)	PGS002769 (Mars 2022, PRS-CS, 1.08 million variants)	1,055,898 1,083,565 (weighted coverage 97%)	5.4	LESS LIKELY	favourable

- **Rheumatoid arthritis:** Already in §18.

Lupus

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Systemic lupus erythematosus	PGS000328 (Reid 2019, 57 variants)	24 57 (weighted coverage 37%)	78.7	TYPICAL	typical
Systemic lupus erythematosus	PGS003960 (Barnado 2023, 57 variants)	56 57 (weighted coverage 99%)	27.3	TYPICAL	typical

- **Systemic lupus erythematosus:** In §18 (76.5 with kNN inference).

Hashimoto's thyroiditis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lymphocytic thyroiditis (Hashimoto)	PGS005270 (White 2025, 55 variants)	50 55 (weighted coverage 70%)	33.8	TYPICAL	typical
Lymphocytic thyroiditis (genomic)	PGS005272 (White 2025, PRS-CS, 1.09 million variants)	1,063,703 1,085,142 (weighted coverage 97%)	51.8	TYPICAL	typical

1 type diabetes

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
1 type diabetes	PGS000021 (Oram 2015, 30 variants)	25 25 (weighted coverage 100%)	41.9	TYPICAL	typical
1 type diabetes (GRS2)	PGS000024 (Sharp 2019, 85 variants (without HLA interactions)	49 53 (weighted coverage 86%)	29.4	TYPICAL	typical

Celiac

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Celiac disease	PGS000040 (Abraham 2014, 228 variants)	122 228 (weighted coverage 50%)	3.9	LESS LIKELY	favourable
Celiac disease	PGS000316 (Sharp 2020, 53 variants)	37 38 (weighted coverage 94%)	31.8	TYPICAL	typical

- **Celiac disease:** In §18 it gave 12.5 with kNN inference; here with the imputation.

Multiple sclerosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Multiple sclerosis	PGS000809 (Barnes 2021, 127 variants)	101 127 (weighted coverage 84%)	23.9	TYPICAL	typical
Multiple sclerosis (genomic)	PGS002726 (Shams 2022, 476 thousand variants)	469,991 476,398 (weighted coverage 98%)	25.1	TYPICAL	typical

- **Multiple sclerosis:** Already in §18.

Vitiligo

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Vitiligo	PGS000738 (Roberts 2019, 48 variants)	46 48 (weighted coverage 90%)	89.9	MORE LIKELY	unfavourable

- **Vitiligo:** In §18 (86.1 with kNN inference).

Spondylitis and psoriatic arthritis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Ankylosing spondylitis	PGS001876 (Privé 2022, 85 variants)	81 85 (weighted coverage 75%)	73.1	TYPICAL	typical
Psoriatic arthritis	PGS000198 (Knevel 2020, 31 variants)	17 18 (weighted coverage 96%)	40.4	TYPICAL	typical

4. Key variants by topic

The most effective HLA markers (read by SNP markers), the genes shared by several autoimmune diseases and the most specific genes of each one.

HLA: the most effective markers

The alleles HLA that give the highest risks (3–50×) of spondylitis, psoriasis, multiple sclerosis, celiac, type diabetes 1 and rheumatoid arthritis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA-B*27 (tag)	rs4349859	B*27	G/G	chip	A (0 copy/s)	favourable	HLA-B27 marker: ~90% of patients with ankylosing spondylitis and anterior uveitis are B27+	Strong
HLA-C*06:02 (tag)	rs12191877	C*06:02	C/C	chip	T (0 copy/s)	favourable	HLA-C*06:02 marker, the greatest genetic factor of psoriasis (~3–4× per copy)	Strong
HLA-C*06:02 (tag 2)	rs10484554		C/C	chip	T (0 copy/s)	favourable	Second marker of HLA-C*06:02 (psoriasis)	Strong
HLA-DRB1*15:01 (tag)	rs3135388	DRB1*15:01	G/G	chip	A (0 copy/s)	favourable	HLA-DRB1*15:01 marker, the largest genetic factor of multiple sclerosis (~3× per copy)	Strong
HLA-DQ2.5 / DR3 (tag)	rs2187668	DQ2.5	C/C	chip	T (0 copy/s)	favourable	DR3-DQ2.5 marker: celiac (~90% of patients), 1 type diabetes, Graves and Hashimoto	Strong
HLA-DQ8 (tag)	rs7454108	DQ8	T/T	chip	C (0 copy/s)	favourable	HLA-DQ8 marker (DR4-DQ8): celiac and diabetes type 1	Strong
HLA-DRB1 (shared epitope)	rs660895		A/A	chip	G (0 copy/s)	favourable	DRB1*04 marker with shared epitope: seropositive rheumatoid arthritis	Strong

Genes shared between autoimmune diseases

T lymphocytes and cytokine signaling are common to several diseases.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PTPN22	rs2476601	R620W	G/G	chip	A (0 copy/s)	favourable	R620W: ~1.5–2 plus rheumatoid arthritis, type 1 diabetes, Graves, Hashimoto, lupus and vitiligo	Strong
CTLA4	rs3087243	CT60	G/A	chip	G (1 copy/s)	unfavourable (light)	CT60: less soluble CTLA-4; thyroid, type 1 diabetes, celiac and rheumatoid arthritis	Moderate
STAT4	rs7574865		G/G	chip	T (0 copy/s)	favourable	The T allele increases the IL-12/IFN signaling: lupus (~1.5) and rheumatoid arthritis (~1.3×)	Strong
TNFAIP3	rs6920220		G/A	chip	A (1 copy/s)	unfavourable (light)	Allele A reduces A20 (NF-κB cross): rheumatoid arthritis, lupus and psoriasis	Moderate
SH2B3	rs3184504	R262W	C/C	chip	T (0 copy/s)	favourable	R262W: celiac, 1 type diabetes and autoimmune hypothyroidism	Moderate
IL2RA	rs2104286		T/T	chip	A (2 copy/s)	unfavourable	IL-2 alpha receptor (CD25): multiple sclerosis and type 1 diabetes [GWAS Catalog: Multiple sclerosis, allele A, OR 1.22, p=2E-47, PMID 24076602]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TYK2	rs34536443	P1104A	G/G	chip	C (0 copy/s)	neutral	P1104A: the C allele (rare) reduces TYK2 and protects from psoriasis, lupus, rheumatoid arthritis and multiple sclerosis	Strong
IL23R	rs11209026	R381Q	G/G	chip	A (0 copy/s)	neutral	R381Q: the A allele (rare) protects from psoriasis, spondylitis and inflammatory bowel disease	Strong

More specific genes of each disease

Lupus (IRF5, ITGAM), spondylitis (ERAP1), type diabetes 1 (INS, IFIH1), multiple sclerosis (IL7R), vitiligo (TYR) and celiac (IL2/IL21).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IRF5	rs2004640		G/G	chip	T (0 copy/s)	favourable	The T allele creates an alternative 1B exon: more interferon type I and ~1.4x plus lupus	Strong
ITGAM	rs1143679	R77H	G/A	chip	A (1 copy/s)	unfavourable (light)	R77H: less functional CD11b integrin; ~1.7x more lupus	Strong
ERAP1	rs30187	K528R	C/C	chip	T (0 copy/s)	favourable	K528R: T allele increases the activity of ERAP1; ankylosing spondylitis in B*27+	Strong
INS (VNTR)	rs689		A/T	chip	A (1 copy/s)	unfavourable (light)	Insulin VNTR marker: less insulin in thymus and less tolerance (1 type diabetes) [GWAS Catalog: Type 1 diabetes, allele A, p=5E-196, PMID 21829393]	Strong
IFIH1	rs1990760		T/T	chip	T (2 copy/s)	unfavourable (light)	A946T: T allele (Thr946) increases MDA5 signaling (viral RNA sensor) and 1 type diabetes (~1.1x); C protects [GWAS Catalog: Type 1 diabetes, allele C, OR 0.89]	Moderate
IL7R	rs6897932		C/C	chip	G (2 copy/s)	unfavourable	T244I: IL-7 receptor; multiple sclerosis [GWAS Catalog: Multiple sclerosis, G allele, OR 1.11, p=2E-8, PMID 21833088]	Moderate
TYR	rs1126809	R402Q	G/G	imputation DR2=1.00	G (2 copy/s)	unfavourable	R402Q: G allele (R402, normal tyrosinase) is associated with more vitiligo; A protects from vitiligo but increases melanoma	Moderate
IL2/IL21 (4q27)	rs6822844		G/T	chip	G (1 copy/s)	unfavourable (light)	IL2/IL21 region: celiac and other autoimmune diseases [GWAS Catalog: Celiac disease, allele G, OR 1.59, p=1E-14, PMID 17558408]	Moderate

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Psoriasis	814 (776)	792 of 1552	776	19.0	favourable
Rheumatoid arthritis	425 (367)	355 of 734	367	61.3	typical
Lupus	432 (343)	282 of 686	341	49.4	typical
1 type diabetes	294 (255)	211 of 510	254	8.5	favourable
Celiac	87 (84)	63 of 168	83	18.0	favourable
Multiple sclerosis	345 (324)	305 of 648	320	21.2	typical
Vitiligo	76 (69)	64 of 138	69	74.2	typical
Ankylosing spondylitis	74 (51)	42 of 102	51	62.6	typical

6. Favourable

- **HLA negative for the most effective 7 markers:** B*27 (rs4349859 G/G), C*06:02 (rs12191877 C/C, rs10484554 C/C), DRB1*15:01 (rs3135388 G/G), DQ2.5 (rs2187668 C/C), DQ8 (rs7454108 T/T), shared epitope (rs660895 A/A). Strong (markers).
- **Rheumatoid arthritis:** percentiles 2.4, 39.2 and 5.4 (seropositive, genomic); PTPN22 G/G, STAT4 G/G. **Moderate to strong.**
- **Psoriasis:** 6.3 and 8.2; loci at 19.0. Psoriatic arthritis at 40.4. **Moderate.**
- **Celiac disease:** 3.9 (50% coverage) and 31.8; loci at 18.0; without DQ2.5 or DQ8, which gives a negative predictive value > 99%. **Strong.**
- **1 type diabetes:** loci at 8.5; scores 41.9 and 29.4. **Moderate.**
- **Multiple sclerosis:** 23.9 and 25.1; loci at 21.2; DRB1*15:01 negative. **Moderate.**
- **Lupus:** complete score at 27.3 and loci at 49.4; IRF5 G/G, STAT4 G/G. **Moderate.**

7. Unfavourable

- **Vitiligo:** percentile 89.9 and loci at 74.2; **TYR R402 G/G** (normal tyrosinase is autoantigen; the same genotype protects from melanoma). **Moderate.**
- **IL2RA rs2104286 T/T** and **IL7R rs6897932 C/C:** 2 risk alleles each (multiple sclerosis and 1 type diabetes, ~1.1–1.2× per copy); **IFIH1 A946T T/T** (1 type diabetes, ~1.1×). Compensated by negative HLA. **Weak.**
- **1 risk allele** in CTLA4 CT60, TNFAIP3, ITGAM R77H (lupus), INS VNTR (diabetes type 1) and IL2/IL21 (celiac). **Weak.**
- **Lupus in the partial coverage score:** percentile 78.7 (37% of the coefficients; the complete score gives 27.3). **Weak.**
- **Ankylosing spondylitis** in 73.1 (typical, without B*27). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	Without the large HLA alleles and without PTPN22 R620W: low tendency for chronic autoimmune inflammation	Favourable
Inflammaging	Psoriasis, low rheumatoid arthritis and celiac arthritis; cytokine loci (TNFAIP3, IL2RA) have a small effect	Favourable
Thymic Involution	Central tolerance depends on thymus (INS VNTR A/T: 1 allele of lower thymic expression of insulin); thymic involution reduces this tolerance with age	Conditional
Loss of T Naive Cells	IL7R C/C and IL2RA T/T: signs of IL-7 and IL-2 that maintain native and regulatory T cells	Neutral
Repertory TCR restriction	HLA class II without risk alleles; the TCR repertoire is restricted with age and favors self-reactive clones	Neutral
Intestinal Barrier dysfunction	Celiac practically excluded (without DQ2.5 or DQ8)	Favourable
Dysbiosis	The fiber-rich microbiota (butyrate) induces regulatory T cells; the more fiber, the better.	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **There is no screening for autoimmune diseases without symptoms.** With this profile, asking for antibodies without symptoms would be mostly false positive. *Suggestion (not applied).*
2. **Celiac:** without DQ2.5 or DQ8, celiac disease is practically excluded (DQ2.2 has not been read). If digestive symptoms arise, look for other causes before cutting gluten. *Suggestion (not applied).*
3. **Vitiligo:** to be attentive to new depigmented spots (hands, face, friction zones). If they appear, go early to the dermatologist, because phototherapy and topical JAK inhibitors work better at the beginning. Sun protection in patches. To ask for TSH (vitiligo is associated with thyroiditis; see §108). *Suggestion (not applied).*
4. **Inflammatory lumbar pain** (morning, which improves with exercise): with negative B*27, spondylitis is unlikely, but not impossible; an MRI of the sacroiliac joints decides. *Suggestion (not applied).*
5. **Already in the plan and protectors:** vitamin D at optimal levels, omega-33.2-4g/day, fiber (the more, the better), zero tobacco and zero alcohol, exercise and sleep. They are the modifiable factors with more evidence in multiple sclerosis, rheumatoid arthritis and psoriasis. *Suggestion (not applied).*
6. **Information:** HLA-B*27 negative and HLA-DRB1*15:01 negative are useful in the clinical record, for example with a uveitis or neurological symptoms. *Suggestion (not applied).*

10. Limitations

- HLA is inferred by marker SNPs, with good but not perfect agreement; clinical HLA typing is the reference standard.
- Some scores have partial coverage (lupus 37%, celiac 50%) and were recalculated with imputation; the values of §18 (kNN inference) are replaced.
- Autoimmune diseases require environmental triggers (infections, tobacco, microbiota, hormones). Genetics alone is not enough to cause them.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published effect allele is compared with the alleles of the imputed VCF. The alleles reported in the strand – were converted and in the palindromic SNPs the strand + (standard of GWAS Catalog and PGS Catalog) was assumed. Alleles with published protective effect ($OR < 1$) were reversed for the sense of risk.
- **Polygenic scores:** GRCh38 harmonized files from PGS Catalog (18 scores in 10 diseases). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 8 diseases (2547 loci), alleles oriented by the sign of the published effect, and empirical calibration in the same reference population.
- **Interpretation:** of Bakker 2006 and Jia 2013 (markers HLA), Monti 2024 and Zhang 2023 (rheumatoid arthritis), Shen 2022 and Saklatvala 2025 (psoriasis), Sharp 2019 (diabetes 1 type), Abraham 2014 and Sharp 2020 (coeliac), IMSCG 2019 (multiple sclerosis), Jin 2016 and Roberts 2019 (vitiligo), Bentham 2015 (lupus), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
HLA: the most effective markers	HLA-B*27 (tag)	rs4349859	G/G	chip	A	0	strand +	without the risk allele
HLA: the most effective markers	HLA-C*06:02 (tag)	rs12191877	C/C	chip	T	0	strand +	without the risk allele
HLA: the most effective markers	HLA-C*06:02 (tag 2)	rs10484554	C/C	chip	T	0	strand +	without the risk allele
HLA: the most effective markers	HLA-DRB1*15:01 (tag)	rs3135388	G/G	chip	A	0	strand +	without the risk allele
HLA: the most effective markers	HLA-DQ2.5 / DR3 (tag)	rs2187668	C/C	chip	T	0	strand +	without the risk allele
HLA: the most effective markers	HLA-DQ8 (tag)	rs7454108	T/T	chip	C	0	strand +	without the risk allele
HLA: the most effective markers	HLA-DRB1 (shared epitope)	rs660895	A/A	chip	G	0	strand +	without the risk allele
Genes shared between autoimmune diseases	PTPN22	rs2476601	G/G	chip	A	0	strand +	without the risk allele
Genes shared between autoimmune diseases	CTLA4	rs3087243	G/A	chip	G	1	strand +	1 copy of risk allele
Genes shared between autoimmune diseases	STAT4	rs7574865	G/G	chip	T	0	strand +	without the risk allele
Genes shared between autoimmune diseases	TNFAIP3	rs6920220	G/A	chip	A	1	strand +	1 copy of risk allele
Genes shared between autoimmune diseases	SH2B3	rs3184504	C/C	chip	T	0	strand +	without the risk allele
Genes shared between autoimmune diseases	IL2RA	rs2104286	T/T	chip	A	2	complement (strand)	2 risk allele copies
Genes shared between autoimmune diseases	TYK2	rs34536443	G/G	chip	C	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Genes shared between autoimmune diseases	IL23R	rs11209026	G/G	chip	A	0	strand +	without the protective allele (common genotype)
More specific genes of each disease	IRF5	rs2004640	G/G	chip	T	0	strand +	without the risk allele
More specific genes of each disease	ITGAM	rs1143679	G/A	chip	A	1	strand +	1 copy of risk allele
More specific genes of each disease	ERAP1	rs30187	C/C	chip	T	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
More specific genes of each disease	INS (VNTR)	rs689	A/T	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
More specific genes of each disease	IFIH1	rs1990760	T/T	chip	T	2	strand +	T/T: 2 copies of the T allele (Thr946), the most risky
More specific genes of each disease	IL7R	rs6897932	C/C	chip	G	2	complement (strand)	2 risk allele copies
More specific genes of each disease	TYR	rs1126809	G/G	imputation DR2=1.00	G	2	strand +	2 risk allele copies
More specific genes of each disease	IL2/IL21 (4q27)	rs6822844	G/T	chip	G	1	strand +	1 copy of risk allele